

Abstract

The purpose of this project is to review Middle East Insurance's portfolio over underwriting years 2021 to 2025 and provide the company with an informed view of its recent performance, outstanding loss obligations, and pricing needs for the 2026 renewal. The dataset includes Medical, Motor, and Property business and reflects results as of 30 September 2025.

During the initial review, we noted that the earlier underwriting years are fully developed, while the more recent ones, 2024 and 2025, are still emerging. The portfolio is also relatively small, which means that individual claims, particularly those with higher severity, can influence results noticeably. This was the case for a large medical claim that affected the more recent years' experience.

The analysis showed stable and profitable outcomes through 2021–2023, followed by a heavier loss year in 2024. By applying monthly incurred development, we estimated a total reserve requirement of roughly \$23,122 for the company. The pricing exercise produced a technical indication of about a 16% rate decrease but given the limited exposure and uncertainty in the latest year, maintaining current rates or applying only a small reduction would be more appropriate.

The project concludes with several practical recommendations, such as closer monitoring of larger claims, improved consistency in case reserving, and offering targeted discounts for low-risk policyholders. These steps aim to support more stable decision-making as the company prepares for the upcoming renewal.

Introduction

Middle East Insurance is a multi-line insurer that has been operating for more than a decade and currently employs around 65 staff members. The company writes business across several lines, mainly Medical, Motor, and Property, and provided detailed data for underwriting years 2021 through 2025 as of 30 September 2025. The objective of this study is to assess the performance of the overall portfolio, evaluate claim development patterns, estimate the required loss reserves, and provide recommendations for the 2026 renewal.

A preliminary review of the dataset showed that the earlier underwriting years are fully developed, while underwriting years 2024 and 2025 remain immature, which introduces additional uncertainty into the analysis. The portfolio itself is relatively small, with exposures ranging from about 8 to 16 Exposure units per year. This limited volume makes the results more sensitive to individual claims, especially those with higher severity, and must be considered when interpreting loss ratios, development patterns, and pricing indications.

Across the portfolio, the experience was generally stable and profitable from 2021 to 2023. However, a noticeable shift occurred in 2024, driven by heavier claim activity and the presence of a significant medical loss that remains partly outstanding. This contributed to higher incurred losses and a sharp increase in the loss ratio for that year. Although 2025 currently shows a more favorable position, the underwriting year has not matured, and further development is expected.

The reserving analysis relied on incurred development using monthly age-to-age factors, which aligned well with the observed claim emergence pattern. The resulting total reserve requirement was estimated at approximately \$23,122, concentrated mainly in the more recent underwriting years. From a pricing perspective, the technical indication suggested a rate decrease of around 16%, but due to limited credibility and recent volatility, a more cautious interpretation was adopted.

This report presents a structured review of the portfolio's performance, a breakdown of results by line of business, development and reserving analyses, pricing indications, and practical recommendations for the company. The aim is to provide Middle East Insurance with an informed and balanced perspective to support renewal decisions and strengthen ongoing portfolio management.

2. Portfolio Overview

2.1 Premium and Claims

UWY	Written Premium	Earned Premium	Paid Claims	Case Reserves	Total Incurred Claims
2021	\$15,050	\$ 15,050	\$ 9,150	-	\$ 9,150
2022	\$27,000	\$ 27,000	\$ 12,650	\$ 2,800	\$ 15,450
2023	\$22,400	\$ 22,400	\$ 11,550	\$ 1,650	\$ 13,200
2024	\$36,600	\$ 36,030.22	\$ 25,850	\$ 7,600	\$ 33,450
2025	\$29,250	\$ 14,987.73	\$ 8,220	\$ 3,100	\$ 11,320

Table 1.1 - Premium and Claims by Underwriting Year

The premium figures show a portfolio that expanded steadily from 2021 through 2024, with written premiums reaching their highest point in 2024 before easing slightly in 2025. Since 2025 is still developing at the valuation date, the lower earned premium is expected. For the first three underwriting years, written and earned premiums match, which indicates that all exposures were fully earned within those years. Small differences appear in 2024 and 2025, mainly due to policies extending into 2026, leaving part of the premium unearned as of September 2025.

Paid losses follow a similar upward pattern and peak in 2024. The absence of case reserves in 2021 shows that all claims for that year were settled. Starting from 2022, the presence of case reserves signals open claims or claims still developing. The higher reserve amount in 2024 reflects the heavier claim activity observed that year. For 2025, both paid losses and reserves are still modest, but this is natural given the immaturity of the underwriting year.

When combining paid losses and reserves, incurred claims remain fairly consistent in the early years before rising sharply in 2024. This makes 2024 stand out as a particularly heavy year, driven by a combination of higher severity and a greater number of open claims. Incurred losses for 2025 are low at this stage, though this does not necessarily reflect its final cost.

Taken together, these observations show a portfolio that grew over the first four years but became more variable, especially in 2024. This growing variability motivates a deeper look into profitability ratios, exposure trends, and claim behavior in the sections that follow.

2.2 Loss and Expense Ratios

UWY	Loss Ratio	LAE Ratio	UW Expense Ratio	Operating Expense Ratio	Combined Ratio
2021	61 %	13 %	24 %	32 %	92 %
2022	57 %	8 %	30 %	34 %	91 %
2023	59 %	8 %	28 %	33 %	92 %
2024	93 %	5 %	22 %	27 %	120 %
2025	76 %	6 %	22 %	27 %	102 %

Table 1.2 - Loss and Expense Ratios by Underwriting Year

The table paints a transparent picture of profitability shifted over the five underwriting years. The first three years maintain consistent loss ratios between 57% and 61%, reflecting steady and predictable claims experience. During this period, LAE ratios remain moderate, and both underwriting and operating expenses stay within expected ranges. As a result, the combined ratios for these years lie below 92%, supporting the view of a stable and profitable book of business.

This condition shifts noticeably in 2024, where the loss ratio rises sharply to 93%. This outcome mirrors the significant increase in incurred losses noted earlier in the premium and claims table. Even though expense ratios show slight improvement, underwriting expenses declining to 22% and operating expenses to 27%, these reductions are insufficient to counterbalance the heavier loss experience. Consequently, the combined ratio escalates to 120%, marking 2024 as an unprofitable underwriting year.

As for 2025, the current loss ratio is 76%, but the underwriting year is still immature, so additional development is expected. Expense ratios remain same as 2024, yet the combined ratio is already above 100%. This early indication suggests that 2025 may also remain pressured as more claims emerge or existing claims develop.

Overall, the ratios display a portfolio that starts from a well controlled and profitable position but increasingly faces pressure in the more recent years. This supports the need to explore frequency, severity, and exposure changes that could explain these emerging patterns.

2.3 Exposure and Claim Measures

UWY	Number of Insured Members	Exposed to Risk Population	Total Number of Claims	Average Premium per ERP	Average Claim per ERP
2021	11	8.02	7	\$ 1,877	\$ 1,141
2022	10	8.02	9	\$ 3,366	\$ 1,926
2023	13	9.68	10	\$ 2,314	\$ 1,363
2024	24	15.93	14	\$ 2,262	\$ 2,100
2025	16	10.76	12	\$ 1,393	\$ 1,052

Table 1.3 - Exposure and Claim Measures by Underwriting Year

The table shows an overall increasing pattern in both the number of insured members and the exposed to risk population from 2021 to 2024, reflecting a steadily expanding book of business. ERP rises from 8.02 in 2021 to 15.93 in 2024. The decline in 2025 is not concerning, as the underwriting year is still immature, and exposure continues to accumulate through the remaining months of coverage.

Total claim counts move in line with exposure, increasing as the portfolio grows and easing slightly in 2025, which is consistent with the current ERP. This behavior is typical for a small book of business where each additional policyholder can noticeably shift claim counts from one year to the next.

Next, average premium per ERP shows some fluctuation across the years. It peaks in 2022 at around \$3,366 and then stabilizes at lower values in the following years. On the other hand, average claim per ERP reaches its highest level in 2024, which aligns with the heavier claim activity observed earlier in the analysis.

One important observation is the small size of the portfolio itself. ERPs range roughly between 8 and 16 units, which means that even a handful of claims, especially medium or large ones, can significantly affect ratios and averages from year to year. This limited volume will play a role later in the report when assessing credibility and interpreting volatility in development patterns, particularly for the more recent underwriting years.

Overall, the exposure measures indicate a growing but still relatively small portfolio where individual claims can meaningfully influence results. This highlights the need for caution when drawing conclusions from short-term fluctuations, especially when the business is still expanding.

2.4 Frequency, Severity, and Loss Cost

The table below summarizes three key indicators of claim behavior:

- **Average claim frequency**, which measures how often claims occur relative to exposure.
- **Average loss severity**, which reflects the average cost per claim.
- **Average loss cost**, which combines both measures to show the expected claim cost per unit of exposure.

UWY	Avg. Claim Frequency	Avg. Loss Severity	Avg. Loss Cost
2021	0.87	\$1,307.14	\$1,141.40
2022	1.12	\$1,716.67	\$1,925.97
2023	1.03	\$1,320.00	\$1,363.33
2024	0.88	\$2,389.29	\$2,099.59
2025	1.12	\$943.33	\$1,052.42

Table 1.4 - Frequency, Severity, and Loss Cost by Underwriting Year

The table shows that claim frequency varies moderately across the years, remaining close to or slightly above one claim per ERP unit. Frequency peaks in 2022 and 2025 at 1.12, while 2021 and 2024 show slightly lower values. Overall, the pattern is relatively stable and does not suggest unusual fluctuation in how often claims occur.

Loss severity, on the other hand, displays more noticeable variation. It reaches its highest level in 2024 at around \$2,389, reinforcing the earlier observation that 2024 experienced heavier claim activity. In contrast, 2025 shows the lowest severity, which is expected for a still developing underwriting year where not all claims have fully matured. The other years fall within a more typical range.

Average loss cost reflects the combined effect of frequency and severity. It increases sharply in 2022 and reaches another peak in 2024, both driven by higher severity. These two years stand out as having the highest cost per unit of exposure. For 2025, the current loss cost is relatively low, but as with the previous tables, this figure is likely to evolve as the underwriting year develops.

As seen, the results highlight the importance of monitoring severity trends as the portfolio expands. While frequency remains fairly consistent, shifts in severity can have a direct impact on underwriting performance and pricing decisions.

3. Loss Ratio by Line of Business

3.1 Medical Line of Business

UWY	Written Premium	Earned Premium	Paid Claims	Case Reserves	Incurred Claims	Loss Ratio
2021	\$ 2,700	\$ 2,700	\$ 2,200	-	\$ 2,200	81 %
2022	\$ 4,450	\$ 4,450	\$ 4,000	-	\$ 4,000	90 %
2023	\$ 8,800	\$ 8,800	\$ 7,000	\$ 700	\$ 7,700	88 %

2024	\$ 15,050	\$14,798.63	\$ 17,150	\$ 4,650	\$ 21,800	147 %
2025	\$ 6,700	\$ 2,583.80	\$ 1,470	\$ 850	\$ 2,320	90 %

Table 2.1 - Medical Line of Business Summary

As seen in the table above, the Medical line shows steady growth in premium from 2021 to 2024, reflecting an expanding presence in this segment. For the first three underwriting years, loss ratios fluctuate between 81% and 90%, which indicates a generally manageable experience with claims that remain in line with earned premium.

The shift in 2024 is immediately noticeable. Incurred losses rise sharply to \$21,800, while earned premium reaches about \$14,799, driving the loss ratio up to 147%, making 2024 a clear outlier. When we investigated the individual claims, the increase was driven not only by higher frequency but also by a few medical cases, one of which continues to develop and was discussed later as part of the outlier review. The presence of a sizeable case reserve in 2024 is consistent with this pattern.

For 2025, the loss ratio returns to 90%, but this figure should be interpreted cautiously. The year has only partially earned premium and several open claims, so additional development is expected. At this stage, the ratio simply reflects the early snapshot rather than the final picture.

When looking at all five years together, the Medical line demonstrates the most pronounced volatility within the portfolio. The spike in 2024, combined with the immature state of 2025, suggests that Medical claims may require closer monitoring going forward, especially where severity is concerned.

3.2 Motor Line of Business

UWY	Written Premium	Earned Premium	Paid Claims	Case Reserves	Incurred Claims	Loss Ratio
2021	\$ 5,050	\$ 5,050	\$ 4,450	-	\$ 4,450	88 %
2022	\$ 4,850	\$ 4,850	\$ 4,250	-	\$ 4,250	88 %
2023	\$ 5,000	\$ 5,000	\$ 3,150	\$ 450	\$ 3,600	72 %
2024	\$ 7,800	\$ 7,482	\$ 3,600	\$ 1,450	\$ 5,050	67 %
2025	\$ 5,150	\$ 2,029	\$ 1,050	\$ 650	\$ 1,700	84 %

Table 2.2 - Motor Line of Business Summary

The Motor line presents a more stable pattern overall. Loss ratios remain at 88% in both 2021 and 2022, reflecting predictable claim activity during these years. In 2023, the ratio improves to 72%, supported by relatively lower incurred claims compared to premium. A similar improvement is seen in 2024, where the loss ratio drops further to 67%. Despite the increase in written premium that year, claims remained well controlled.

In 2025, the loss ratio increases to 84%. Since the year is still immature and only part of the premium has been earned, this result should be interpreted with caution. Additional claim development may shift the ratio as the year matures.

Compared to the Medical line, Motor results show less volatility and a more consistent trend over time. The improvements in 2023 and 2024 suggest favorable performance, even though the early indication for 2025 shows a slight reversal.

3.3 Property Line of Business

UWY	Written Premium	Earned Premium	Paid Claims	Case Reserves	Incurred Claims	Loss Ratio
2021	\$ 7,300	\$ 7,300	\$ 2,500	–	\$ 2,500	34 %
2022	\$ 17,700	\$ 17,700	\$ 4,400	\$ 2,800	\$ 7,200	41 %
2023	\$ 8,600	\$ 8,600	\$ 1,400	\$ 500	\$ 1,900	22 %
2024	\$ 13,750	\$ 13,750	\$ 5,100	\$ 1,500	\$ 6,600	48 %
2025	\$ 17,400	\$ 10,375	\$ 5,700	\$ 1,600	\$ 7,300	70 %

Table 2.3 - Property Line of Business Summary

The Property line is generally the most stable and profitable segment in the portfolio. Loss ratios remain low from 2021 through 2024, ranging from 22% to 48%. These values indicate relatively light claim activity for this line, even in years where premium or exposure increased. The presence of small case reserves between 2022 and 2024 reflects normal ongoing development but does not signal any major issues.

In 2025, the loss ratio increases to 70%. As with the other lines, this figure is influenced by the immaturity of the underwriting year—earned premium is still accumulating, and several claims remain open. Given the historical pattern of this line, it is too early to treat the 2025 ratio as a sign of deterioration.

When comparing the three lines of business, the differences become clear. The Medical line shows the most variability, largely due to higher-severity claims. The Motor line remains relatively stable with gradual improvements in recent years. The Property line has been consistently profitable, contributing positively to the overall portfolio's balance.

These contrasts help explain the shifts observed in the consolidated results and provide a clearer view of which segments are driving recent performance trends.

4. Data Review and Outliers Overview

4.1 Data Quality and Consistency Review

Before proceeding with the development analysis, we reviewed the dataset to make sure the information was consistent across policies, loss dates, exposures, and incurred amounts. Overall, the data structure is clean and follows expected patterns. Policy effective dates align properly with underwriting years, and reported losses occur within the policy coverage periods. We did not encounter cases where claims fell outside the insured timeframe.

Underwriting years 2021 to 2023 appear fully mature: exposures are complete, and there are no outstanding case reserves. On the other hand, UWYs 2024 and 2025 are still developing as of the valuation date (30 September 2025), which explains the partial exposures and the presence of open claims. These later years naturally carry more uncertainty and will require development adjustments when estimating ultimate losses.

We also noticed some claims with a recorded loss date but no payment and no outstanding reserve. After looking closer, these appear to be reported claims that were either withdrawn, fell below a deductible, or concluded without cost. Since the dataset does not provide additional claim details, treating them as zero-incurred events was the most reasonable and internally consistent choice for a portfolio of this size.

Exposure values behave as expected based on policy terms and valuation timing. Partial exposure for policies extending into the following year is handled correctly, and the earned premium amounts reflect this pattern. Nothing in the exposure fields suggested systematic errors.

Overall, the dataset is suitable for development analysis. The main considerations we kept in mind are the immaturity of the more recent underwriting years, the relatively small size of the portfolio, and the presence of some zero-cost reported claims. These factors became important when interpreting volatility in the development patterns later in the study.

4.2 Outliers Overview

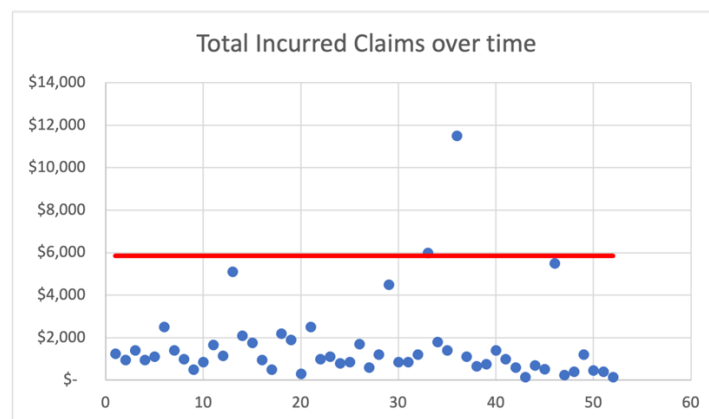


Figure 1 - Incurred Claims with 97.5th Percentile Threshold Line

While reviewing the data for outliers, we checked for unusually large severities using the 97.5th percentile threshold, which came out to \$5,863. Two incurred claims exceeded the threshold, a Property claim around \$6,000 and a Medical claim of \$11,500.

The Property claim was close enough to the threshold that we felt comfortable keeping it fully in the dataset, since similar severities appeared elsewhere in the portfolio and removing it would have distorted the representation of Property risks.

On the other hand, the Medical claim was different. It was the only case of that magnitude across the four years and stands out as an extreme value, especially given the small size of the portfolio. In a real setting, we would ask for medical reports or discuss the case with the claims department to confirm whether it reflects an isolated event or exposure that could recur. Since such details were not available, we did not want this single claim to drive the development factors too strongly.

Since such an extreme event appears once in four years, a weight of 25 % was applied when incorporating it into the mean-based factor selection. This allowed us to acknowledge the claim without letting it distort the overall pattern, which felt like the most balanced approach for a small portfolio with limited data.

In general, the dataset does not contain many high severity cases, so even one large claim can noticeably affect the development pattern. Accounting for this appropriately helps avoid overstating future losses, especially for UWYs 2024 and 2025.

With the dataset reviewed and the outlier treatment decided, we proceeded to build the monthly development triangles and estimate ultimate losses for the more recent underwriting years.

5. Loss Development and Ultimate Loss Estimation

After reviewing and preparing the dataset, we moved into the development analysis to estimate ultimate losses for UWY 2025. Although the original requirement suggested working with quarterly development, we quickly realized that quarterly aggregation was not ideal for this portfolio. With such a small number of claims, several payments occurred within the same month, and grouping them quarterly tended to smooth out meaningful variation.

Therefore, we worked with monthly incurred development, which provided a clearer view of how losses emerged over time across all underwriting years.

5.1 Building the Monthly Development Triangles

The monthly incurred development triangles, provided in the appendix, show how claim values evolved from early reporting stages to maturity for UWYs 2021 through 2025. Reviewing the five years history shows that claims generally stop developing beyond 23 months. This was consistent across the older underwriting years, which helped validate the assumption of a roughly two-year development horizon for this portfolio.

For the mature years (2021-2023), most claim activity concentrated within the first several months, with some adjustments continuing into the second year. This is typical of shorter tailed lines, where claims settle relatively quickly. Meanwhile, UWYs 2024 and 2025 had not yet completed the full development span, so we relied on historical patterns to complete their development.

The triangles confirmed the portfolio's overall behavior, rapid early emergence, moderate adjustments afterward, and stabilization around the two-year mark. This helped guide our selection of development factors.

5.2 Age-to-Age Factors Selection

Age-to-age factors measure how incurred losses evolve from one development period to the next and form the core of the ultimate loss projection. To derive credible factors, we calculated several averaging methods, simple average, volume-weighted mean, 3-year average, and geometric mean. UWY 2025 was excluded from the derivation because its early development is too unstable to provide reliable information.

Each averaging method was evaluated by testing its ability to project incurred losses from January 2025 to September 2025. The simple and 3-year averages were inflated by the heavy losses observed in 2024, while the geometric mean smoothed variations but understated the early emergence typical of this portfolio.

The volume-weighted mean produced the closest projection (\$12,275 vs. actual \$11,320) and was therefore selected as the primary basis for factor selection.

Across all underwriting years, most first claims were reported within the first three months, and development during this period was minimal. Assigning higher factors here would introduce artificial volatility, so the 2/1 and 3/2 factors were conservatively set to 1.

During the reasonableness checks, two unusually high factors, around 5, appeared in the 4/3 column, resulting in an average of 2.55. These arose from very small initial incurred amounts followed by normal subsequent development. Reducing these factors to 2 underestimated the development in earlier years, confirming that the high values reflect true volatility rather than data errors, and therefore, they were retained.

We also examined the previously discussed claim outlier of \$11,500. To assess its influence, we constructed an adjusted triangle where this claim was replaced with 25% of its value. New age-to-age factors were calculated beginning from the 9/8 column, as the outlier emerged at month 9. The volume-weighted mean remained the most appropriate method under this adjusted scenario as well.

The main triangle continues to reflect the real experience of the portfolio, so the outlier was kept in the primary data. The adjusted triangle was used only to evaluate the sensitivity of factor selection.

Development Year 0	2/1	1	Tail factor 1/12	1.097	Development Year 1	2/1	1
	3/2	1				3/2	1
	4/3	2.550				4/3	1.017
	5/4	1.441				5/4	1
	6/5	1.310				6/5	1
	7/6	1.519				7/6	1.030
	8/7	1.168				8/7	1
	9/8	1.185				9/8	1
	10/9	1.154				10/9	1
	11/10	1.030				11/10	1.030
	12/11	1.112				12/11	1

Table 3.1 - Age-to-age selected incurred losses development factors

To validate the factor selections, we reapplied the final development factors to the January 2025 incurred losses. The projection yielded \$10,119 compared with the actual \$11,320. This slight difference is reasonable given the unusually rapid claim emergence early in 2025

Next step was estimating the ultimate losses for UWY 2025 under each averaging method:

2025 Ultimate losses	
Average	\$ 22,843
Mean (outlier included)	\$ 16,658
3-year Average	\$ 21,908
Geometric Mean	\$ 20,626
Selected Mean	\$ 17,723

Table 3.2 - 2025 Estimated Ultimate losses under different averages

Based on the selected development factors, the technical estimate for the ultimate loss of UWY 2025 is \$17,723.

Since 2025 is still immature and the portfolio is small, some uncertainty remains around late reported claims. To account for this uncertainty, without overstating it, we applied a 5% prudence margin, aligned with common practice in small sample reserving.

This brought the final selected ultimate loss for 2025 to \$18,289.

The value sits above several earlier underwriting years, which is consistent with the increased volatility seen recently, without assuming if the unusually heavy losses in 2024 will necessarily repeat.

6. Ratemaking Indication for UWY 2025

Based on the trended and developed incurred losses for underwriting year 2025, we estimated the expected cost of 2025 claims at \$18,289. The company's earned premium at current rates for the same year is \$29,000, and the company's permissible loss ratio is 75%.

Using this information, we calculated the indicated change to existing rates under both the loss ratio method and the loss cost method.

Although only the loss ratio method was required, we used the second method as a cross-check to make sure our results were consistent.

6.1 Loss Ratio Method

Under the loss ratio approach, we first calculated the expected effective loss ratio implied by the trended and developed incurred losses:

$$\begin{aligned}\text{Expected Effective Loss Ratio} &= \frac{\text{Estimated Ultimate losses}}{\text{Earned Premium at current rates}} \\ &= \frac{18,289}{29,000} \\ &= 63 \%\end{aligned}$$

This effective Loss Ratio is lower than the permissible Loss Ratio of 75%, meaning the portfolio is performing better than the target level.

The indicated rate change therefore is:

$$\begin{aligned}\text{Indicated rate change} &= \text{Expected Effective Loss Ratio} - \text{PLR} \\ &= 63 - 75 \\ &= -16 \%\end{aligned}$$

The resulting percentage suggests a decrease in rates. Another way to think about it is to compute the premium that would match the target loss ratio:

$$\text{Indicated Premium} = \frac{18,289}{0.75} = \$ 24,385$$

which is lower than the current earned premium of \$29,000. In other words, based solely on the 2025 experience and the selected PLR, current rates appear more than adequate.

While the math is straightforward, we kept in mind throughout the report that 2025 is a small and immature underwriting year. Thus, we later toned down the final pricing recommendation despite the technical result.

6.2 Loss Cost Method

Even though the project did not require a loss cost approach, we used it as a secondary check.

Using the earned exposure of 10.756 units, the new average loss cost implied by the trended and developed incurred losses is:

$$\text{New Average Loss Cost} = \frac{18,289}{10.756} = \$ 1,700$$

To compare this to current pricing, we calculated the new average gross rate:

$$\text{New Average Gross rate} = \frac{1,700}{0.75} = \$ 2,267$$

and the current average gross rate based on existing premiums:

$$\text{Current Gross Rate} = \frac{\text{Earned Premium at current rate}}{\text{Earned Exposure units}} = \frac{29,000}{6.81} = \$ 2,696.13$$

Comparing the two gross rates values:

$$\text{Indicated Rate Change} = \frac{2,267 - 2,696.13}{2,696} = -15.9 \%$$

The loss cost method therefore confirms the result obtained from the loss ratio method decrease of approximately 16%.

Both approaches therefore lead to the same conclusion, that from a pure objective view, the technical indication points to a moderate rate decrease.

However, because the 2025 underwriting year has limited exposure and is still developing, we treated this indication cautiously in the recommendations section.

7. Loss Reserve Estimation

7.1 Choice of Reserving Method

To estimate the loss reserves for this portfolio, we selected the incurred Chain Ladder method as the primary approach. Before finalizing this choice, we considered alternative methods and evaluated how well each one aligned with the characteristics of the dataset.

The Loss Ratio method was not retained since it relies mainly on premiums and expected loss ratios instead of the actual development of claim costs. For a portfolio of this size, where individual claims can materially shift results from one year to the next, such an approach would introduce unnecessary noise.

We also examined the Bornhuetter-Ferguson method, which blends expected losses with observed development. While the project provides a permissible loss ratio of 75%, this figure reflects the company's pricing target, not an objective expected loss ratio. Using the PLR as a substitute for an ELR would therefore introduce bias especially for a small portfolio where credibility is limited, and recent experience has been volatile. Without a reliable data or credible external benchmark, applying BF would make the reserve more dependent on assumptions than on observed claim behavior.

In contrast, the incurred Chain Ladder method relies directly on historical development patterns within the portfolio. The incurred triangles show a clear stabilization around 23 months, and case reserves appear to be updated in a timely manner. These features make incurred development a reasonable and transparent basis for projecting ultimate losses, particularly when combined with sensitivity checks around outliers and recent volatility.

For these reasons, and after reviewing the alternatives, the incurred Chain Ladder method provided the most defensible and internally consistent reserve estimate for this assignment.

7.2 Projection of Ultimate Incurred Losses

Using the previously derived monthly development factors, we projected the ultimate incurred losses for UWYs 2024 and 2025. Underwriting years 2021 to 2023 were already fully developed, so their ultimate values are simply their incurred amounts to date.

The projected ultimate losses were:

- **2021:** \$ 9,150
- **2022:** \$ 15,450
- **2023:** \$ 13,200
- **2024:** \$ 34,453
- **2025:** \$18,892

These results reflect both the maturity of older underwriting years and the increased volatility in more recent years, particularly 2024 and 2025.

7.3 Paid Development Analysis (Reasonableness Check)

As a validation step, we built monthly paid loss development triangles and calculated the corresponding age-to-age paid development factors. Paid losses in this portfolio develop more slowly than incurred losses, due to the timing of claim payments and the early establishment of case reserves.

When we tested these factors by projecting January 2025 paid losses forward, the estimates ranged from about \$4,381 to \$14,729, depending on the averaging method. These projections either underestimated or over expected the actual incurred amount of \$11,320 at September 2025.

This underestimation made sense after we reviewed the claim files. The portfolio uses case reserving actively, and payments often lag incurred recognition, especially for the larger or more complex claims. Because of this, paid development alone does not reliably capture the true emergence pattern in this dataset.

For that reason, we used the paid Chain Ladder results only as a reasonableness check and kept the incurred Chain Ladder method as our primary reserving tool.

7.4 Case Reserve Determination and Total Reserve Estimate

With the ultimate incurred losses determined, we calculated the following reserve components for each underwriting year:

- $\text{Case Reserve} = \text{Incurred Losses to Date} - \text{Paid Losses to Date}$
- $\text{IBNR} = \text{Ultimate Losses} - \text{Incurred Losses to Date}$
- $\text{Total Loss Reserve} = \text{Case Reserve} + \text{IBNR}$

Older underwriting years (2021-2023) are essentially fully developed, with only minor outstanding case reserves in 2022 and 2023. Most of the reserving need lies in the recent years, which is consistent with their immaturity and higher volatility.

UWY	Ultimate Losses	Paid-to-date Losses	IBNR	Case Reserve	Total Reserve
2021	\$ 9,150	\$ 9,150	-	-	-
2022	\$ 15,450	\$ 12,650	-	\$ 2,800	\$ 2,800
2023	\$ 13,200	\$ 11,550	-	\$ 1,650	\$ 1,650
2024	\$ 34,453	\$ 25,850	\$ 1,003	\$ 7,600	\$ 8,603
2025	\$ 18,289	\$ 8,220	\$ 6,969	\$ 3,100	\$ 10,069
Total	\$ 90,542	\$ 67,420	\$ 7,972	\$ 15,150	\$ 23,122

Table 4 - Estimated Loss Reserves

Thus, the total required reserve for the portfolio is of \$23,122, consisting of \$7,972 in IBNR and \$15,150 in case reserves.

Consideration of the Medical Outlier

One high-severity medical claim remains outstanding and has materially influenced the development pattern, particularly in the later months of UWY 2024 and 2025. Although its impact is appropriately captured in the incurred development factors, its partially paid status introduces additional uncertainty.

From a reserving and risk-management perspective, this claim should be monitored proactively, with periodic status updates, medical progress evaluations, and potential future cost assessments. Ensuring early recognition of any deterioration in this case will help maintain reserve adequacy and reduce the risk of sudden reserve strengthening in future reporting periods.

8. Analysis and Interpretation of Results

8.1 Overall Interpretation

When we brought together the results from the reserving and pricing analyses, a clearer picture of the portfolio's recent behavior started to emerge. The mature underwriting years show predictable development patterns and stable incurred losses. Their claim activity settles in a way that aligns well with short, tailed business, which increases our confidence in the development factors derived from those periods.

The more recent underwriting years behave differently. In both 2024 and 2025, losses emerged faster and with noticeably more variation. Part of this comes from the presence of higher severity claims, including the large medical case that remains open. Another factor is the small size of the portfolio: with exposures ranging from 8 to 16 ERP units, even one or two sizeable claims can materially shift the results.

Because of this, the uncertainty surrounding the recent years should not be interpreted as a structural change in risk without further evidence. Instead, it reflects the natural volatility that comes with limited data and the fact that 2024 and 2025 have not yet fully matured.

From a reserving perspective, the total requirement of \$23,122 reflects the combination of stability in older years and the heightened uncertainty in the most recent ones. The distribution of reserve amounts minimal in the early years and concentrated in 2024 and 2025, is consistent with normal claim development and aligns with what we observed in the triangles.

On the pricing side, both methods pointed to an indicated decrease of around 16%. While the math behind this result is sound, our team discussed whether it would be appropriate to act directly on such a strong indication. Because the 2025 underwriting year has very limited exposure and is still developing, its experience should not be given full weight. The contrast between the heavy losses in 2024 and the lighter results in early 2025 also shows how sensitive results are to timing and emergence. That sensitivity reduces the credibility of a single-year indication.

Overall, while the portfolio remains operationally sound, the recent volatility suggests that future decisions, whether related to pricing, monitoring, or reserving, should be taken with some caution, particularly for segments with higher severity risk.

8.2 Recommendations

Based on the analysis, several practical recommendations can help guide the company's next steps.

1. Adopt a conservative pricing position for 2026.
Even though the technical indication points to a 16% rate decrease, we do not believe the recent experience carries enough credibility to support such a large adjustment. Given the immaturity of 2025 and the volatility seen in 2024, a more balanced approach would be to keep current rates stable or consider at most a small decrease of up to 5%. This acknowledges the technical result without overreacting to short-term fluctuations.
2. Strengthen monitoring of higher severity or longer duration claims.
The medical outlier highlighted in the analysis had a noticeable impact on development patterns. Since it remains partially unpaid, it should be followed closely. Regular updates on case status, treatment progress, and potential future costs would help prevent unexpected reserve increases and improve the accuracy of future projections.
3. Review and formalize case reserving practices.
The comparison between paid and incurred development suggests that case reserving plays a significant role in how losses emerge. Ensuring that case reserves are set on a consistent and well-documented basis, especially for complex or longer tail claims, would reduce unnecessary volatility in the incurred development.
4. Use targeted pricing actions instead of broad rate cuts.
Because the portfolio is small, year-to-year results will continue to fluctuate. Instead of reducing rates across the entire book, the company could offer selective discounts or promotional incentives for lower-risk policyholders. For motor, discounts could be directed

toward safe drivers or policyholders in defensive-driving programs. Property discounts should remain limited to clearly low-risk homes to avoid weakening the strong performance of that line. This allows the company to remain competitive and attract desirable risks without compromising overall rate adequacy.

5. Continue monitoring exposure growth and its impact on credibility.

As the portfolio grows, long-term patterns will become clearer and technical indications will carry more weight. In the meantime, decisions should continue balancing technical results with practical judgment, especially when exposure remains small.

Overall, these recommendations aim to help the company maintain financial stability, manage uncertainty in recent experience, and gradually strengthen the credibility of its data as the portfolio evolves.

Conclusion

The study provided a comprehensive view of how losses develop within the portfolio and how these patterns influence both reserving and pricing decisions. The mature underwriting years behaved predictably, while the more recent years showed faster and more variable emergence, partly driven by higher-severity claims. The final reserve estimate of \$23,122 reflects this combination of stability and uncertainty.

On the pricing side, the technical indication supported a notable decrease. However, given the small exposure base and the immaturity of 2025, relying entirely on that indication would be premature. A more careful approach, maintaining current rates or applying only a modest reduction, is more appropriate at this stage.

Moving forward, closer monitoring of developing claims, especially those with higher severity, along with consistent case reserving practices, will help the company manage volatility and maintain stable projections. These actions will support more reliable underwriting and reserving outcomes as the portfolio continues to grow.